Including ancestral state estimation uncertainty can improve disparity analyses

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Abstract

Morphological trait datasets and phylogenies are routinely paired to investigate macroevolutionary patterns during disparity analyses. However, incomplete fossil sampling can distort disparity estimates, obscuring true evolutionary signals. Ancestral state estimation can be used for both continuous and discrete traits to extend these analyses beyond incomplete fossil data and generate a more complete trait space. However, when ancestral state estimation occurs in the disparity pipeline, and the inevitable uncertainty in these estimates, complicate their integration. Determining the most robust workflow for integrating ancestral state estimation in disparity analyses remains a critical methodological challenge. Using simulations to attain a ground-truth disparity value, we evaluated different approaches to performing ancestral state estimation and incorporating uncertainty across varying continuous and discrete trait models, fossil sampling densities and disparity metrics. Ancestral state estimation generally improved recovery of true disparity relative to tip-only analyses, though the optimal approach depended on the interaction between trait model and fossil sampling density. For continuous traits, probabilistic approaches were most accurate, but were sensitive to model misspecification under low fossil sampling density. For discrete traits, pre-ordination methods were most reliable and probabilistic approaches outperformed point estimates under low sampling, while point estimates became increasingly accurate as sampling density increased. Fossil sampling density was a stronger predictor of disparity accuracy than estimation method choice, underscoring that methodologies are only as powerful as the data provided. Our findings offer a practical decision framework for selecting the most appropriate workflow given the sampling density and trait characteristics of a dataset.

Keywords: ancestral state estimation; disparity; fossils; simulation; uncertainty; macroevolution

1 Introduction

Morphological diversity, hereafter termed disparity, is widely used for investigating macroevolutionary dynamics (Foote 1997; Wills 2001; Guillerme et al. 2020a). Disparity can be described as the distribution or spread of taxa within a multidimensional trait space defined by morphological similarity (Erwin 2007) and can be quantified using both continuous (García-Navas et al. 2017; Gray et al. 2019; Friedman et al. 2020) and discrete characters (Foote 1994; Lloyd 2016). Morphological trait datasets are routinely integrated with phylogenies when examining disparity across clades or time. These analyses can be used to investigate selectivity across mass extinctions (Bapst et al. 2012; Cole and Hopkins 2021; Liu et al. 2024), reveal patterns in morphological convergence (Muschick et al. 2012; Stayton 2015, 2024) or characterise the tempo and mode of morphological diversification during adaptive radiations (Harmon et al. 2003; Mahler 2010; Hughes et al. 2013). However, such analyses are inevitably limited by the incompleteness of the fossil record; only a biased subset of past diversity is preserved, leaving phylogenetically-implied ghost taxa and obscuring large portions of a clade's evolutionary history (Norell 1993; Wagner 1995; Flannery Sutherland et al. 2019; Smith et al. 2021). These missing data can result in both under- and over-estimations of disparity (Mitchell 2015; Smith et al. 2023).

One way in which unsampled lineages can be accounted for is through the use of ancestral state estimation. Ancestral state estimation is the method of estimating trait value(s) in a set of common ancestors, i.e., nodes on a phylogenetic tree, given a set of tip data (Schluter et al. 1997; Cunningham et al. 1998; Revell 2025). These estimated states can be combined with the observed tip data to generate a more complete trait space, filling the gaps left by an incomplete fossil record and, in theory, improving our ability to recover the 'true disparity' of a group (Sidlauskas 2008; Brusatte et al. 2011; Lloyd 2018). True disparity in this context can be defined as the total trait space occupied by all lineages throughout a clade's history, measured by either volume or density (Guillerme et al. 2020b). Crucially, the inclusion of ancestral states allows disparity through time analyses to be explored with higher temporal resolution (Brusatte et al. 2011; Halliday and Goswami 2016; Guillerme 2018).

Despite the aforementioned advantages, the use of ancestral state estimation within disparity analyses has often been neglected. The hesitancy to incorporate ancestral state estimation has likely stemmed from the notion that estimates are inherently uncertain; root-ward estimates in particular are often associated with large confidence intervals or inconclusive scaled likelihoods

(Schluter et al. 1997; Losos 1999; Gascuel and Steel 2014). Despite the encouragement to propagate uncertainty during comparative analyses (Silvestro et al. 2015; Boyko and O'Meara 2024), ancestral state estimations are generally treated as point estimates during disparity analyses, because there is little guidance on how to correctly treat ancestral state estimation uncertainty downstream when constructing the trait space. These point estimates are typically the maximum likelihood estimates for continuous characters or collapsed states returned from arbitrary thresholds for discrete characters (Smith and Donoghue 2022; Casali et al. 2023; Li et al. 2026). By ignoring ancestral state estimation uncertainty, point estimates can lead to overconfidence in results and incorrect inferences in downstream analyses, all while obscuring potentially accurate signals (Ackerly et al. 2006; Vanderpoorten and Goffinet 2006).

Another key issue with ancestral state estimation is its sensitivity to model misspecification. Likelihood-based approaches to ancestral state estimation require both a tree topology and a specified model of character evolution. Continuous character estimations are typically modelled under Brownian motion (BM; but see Elliot and Mooers 2014; Castiglione et al. 2020; Meade and Pagel 2022), which describes a directionless, stochastic process where the trait divergence between related taxa is proportional to their shared evolutionary history and the rate of evolution (Felsenstein 1985; Schluter et al. 1997; O'Meara et al. 2006). However, deviations from BM in the trait data, such as directional trends or stabilising selection, inevitably returns biased estimates (Webster and Purvis 2002; Finarelli and Flynn 2006; Albert et al. 2009; Finarelli and Goswami 2013). Model-related issues can also arise for discrete traits, which are estimated using Mk (Pagel 1994, 1999; Lewis 2001) or SSE models (Ng and Smith 2014; Holland et al. 2020). Estimation accuracy of discrete states appears prone to model over-parameterisation and sensitive to fast transition rates (Mooers and Schluter 1999; Keating 2023). Furthermore, discrete ancestral state estimation faces a paradoxical trade-off where a slow rate of evolution allows for an accurate estimation of the root state, but provides little information for calculating transition rates (Gascuel and Steel 2020). Similarly, the opposite is true whereby fast rates allow for an accurate estimation of transition rates, but uncertainty in the root state estimation.

Given that uncertainty is often high in ancestral state estimation, how then can estimates be improved? Ancestral state estimation is statistically inconsistent, meaning that even if the sample size of extant tips were infinite, estimates are not expected to converge on the true value (Ané 2008). The inclusion of fossils can improve estimations by breaking long branch lengths, reducing distance between tips and internal nodes (Finarelli and Flynn 2006; Finarelli and Goswami 2013;

Joy et al. 2016; Goswami and Clavel 2025). Numerous simulation and empirical based studies have shown that ancestral state estimations based on extant-taxa only are biased (Webster and Purvis 2002; Finarelli and Flynn 2006; Albert et al. 2009; Betancur-R et al. 2015; Royer-Carenzi and Didier 2016; Monson et al. 2022; Wisniewski et al. 2022; Keating 2023), though this bias can be driven by model misspecification rather than missing data alone (Goldberg and Igić 2008; Puttick and Thomas 2015). Similar biases have been observed in extant-only disparity analyses (Mitchell 2015). Despite this, many studies are forced to use extant-only trees for both ancestral state estimation (Jetz and Pyron 2018; Amador and Giannini 2021) and disparity (Harmon et al. 2010; Baker et al. 2015; Smith and Donoghue 2022) due to a lack of viable fossil data.

Numerous studies have assessed the raw estimation error of ancestral state estimation approaches under different models of evolution, methods of estimation and fossil sampling density (Elliot and Mooers 2014; Royer-Carenzi and Didier 2016; Reyes et al. 2018; Holland et al. 2020; Keating 2023; Boyko et al. 2026). However, how trait estimation accuracy is propagated into the recovery of an accurate trait space is less understood, despite the continued use of ancestral state estimation in disparity analyses. One methodological debate concerns whether ancestral state estimation should be conducted before or after trait space ordination. Previous work has found that carrying out ancestral state estimation on the axes of an ordination matrix rather than raw characters can distort the distribution of points in a trait space markedly, introducing unwanted phylogenetic signal and low levels of convergence (Lloyd 2018). However, this study was restricted to a single empirical dataset and discrete traits. Thus, we lack a comprehensive understanding of how estimation approaches, trait data and fossil sampling density interact to affect our ability to recover accurate patterns in trait space and therefore disparity.

Here we evaluate the efficacy of different ancestral state estimation approaches in recovering true disparity values across different evolutionary models and fossil sampling densities using a simulation framework. Simulations offer a robust approach for benchmarking methodologies because a ground-truth dataset can be generated, so trait values and the resulting disparity value are all known *a priori* (Lotterhos et al. 2022). Specifically, we directly compare pre- versus post-ordination ancestral state estimation approaches, as well as point-estimate versus probabilistic treatments of uncertainty. We also investigate how estimation error varies under different models of continuous (Brownian motion with and without trend and Ornstein-Uhlenbeck) and discrete traits (varying transition rates), fossil sampling densities (100%, 50%, 15%, 5% and 0% of extinct lineages) and disparity metrics examining size and density. This integrative approach provides a

comprehensive evaluation of when ancestral state estimation improves disparity estimation, when it introduces bias, and how uncertainty should be treated when working with an incomplete fossil record.

2 Materials and Methods

2.1 Simulation Framework

2.1.1 Phylogenies

All simulations were generated using `treats` in R v.4.4.1 (Guillerme 2024; R Core Team 2025). We simulated birth-death trees with the speciation (λ) parameter set to 1.0, and extinction (μ) set to 0.9 (Figure 1 Step 1). This fraction of extinction ($\mu = 0.9$) produces trees that match our expectation of substantial extinct diversity based on the fossil record (Scholl and Wiens 2016; Pires et al. 2018; Henao Diaz et al. 2019; Morlon et al. 2024). We generated three sets of 100 birth-death trees with these parameters until the simulations reached 50, 100 or 150 extant taxa, with maximum tree size of 500, 1000 and 1500 tips, respectively (Figure S1). The randomness of the birth-death model meant some simulated trees contained only extinct taxa, so we re-ran each individual simulation until we obtained the desired number of extant taxa. This meant that although the parameters were set to $\lambda = 1$ and $\mu = 0.9$, the effective parameters slightly deviated from these input values (Figure S2). Mean net diversification across all simulated trees was 0.30 ± 0.09. Constant birth-death parameters meant that the simulated trees were highly symmetrical (Figure S3; Khurana et al. 2024). Given that empirical phylogenies tend to be imbalanced (Mooers and Heard 1997; Harcourt-Brown et al. 2001), we simulated 100 birth-death trees with maximum tree size of 50 extant and 500 total tips with high asymmetry and ran the same pipeline (Figure S4, Figure 1). Trees were pruned to extract the crown clade of the tree for downstream analyses. This ensured that all downsampled trees shared an identical root node and tree height with the original tree simulation. Therefore, any tip included in the analyses was a descendant of the most basal node considered, i.e., the MRCA of extant tips, ensuring that all ordinations and disparity values were calculated within a consistent clade.

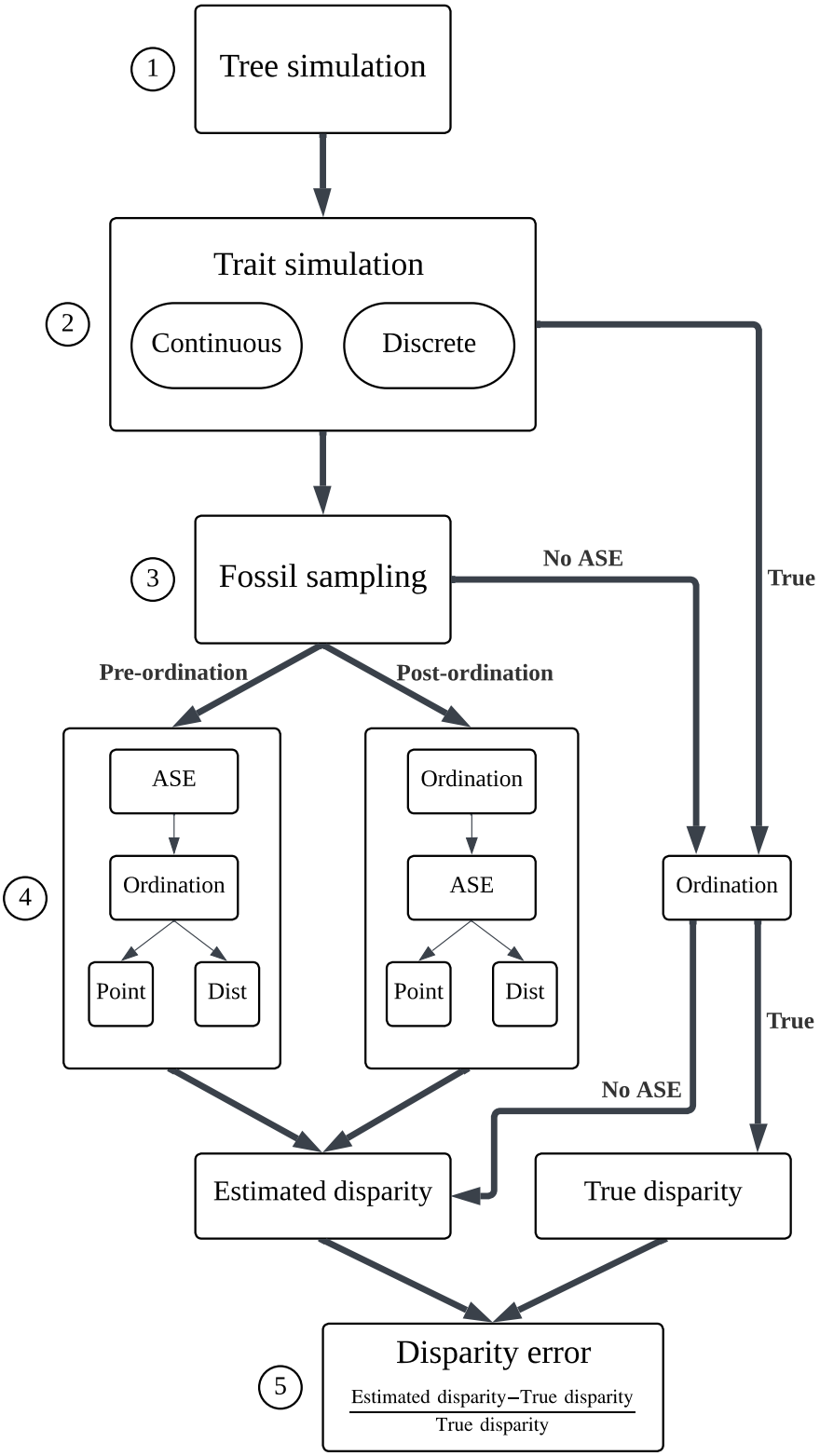


Figure 1: Overall simulation pipeline, highlighting the five different estimation routes (pre-ordination point, pre-ordination distribution, post-ordination point, post-ordination distribution and no ancestral state estimation) as well as the true route, from which ground-truth disparity value is calculated from the untreated simulation.

2.1.2 Trait simulations

For each phylogeny, we simulated continuous and discrete trait data separately (Figure 1 step 2). Continuous trait data were simulated under variations of Brownian motion (BM) and Ornstein-Uhlenbeck (OU) models (Table 1). We simulated BM with variance = 0.25. We also simulated BM with a trend parameter of 0.3, by which the ancestral trait value increased incrementally proportional to branch length. OU trait data were simulated with the α parameter set based on phylogenetic half-life to account for differing tree heights across simulations (Gearty et al. 2024).

$$\alpha = \frac{\ln(2)}{\text{tree height} \times 0.25} \tag{1}$$

We simulated 20 independent characters for each model (BM, BM + trend and OU) across each phylogeny.

Discrete trait evolution was simulated using a continuous-time Markov process, whereby the probability of state transition along each branch was proportional to the product of the transition probability and branch length. We simulated discrete traits across three rate categories, representing fast (0.9), medium (0.1) and slow (0.01) transition rates (Table 1). These rates were assigned by generating equal-rates (ER) transition probability matrices. For each category, 85 binary characters and 15 three-state characters were simulated, following a typical scoring ratio found in empirical data (Guillerme and Cooper 2016).

Table 1: Simulation models and parameters used for phylogeny generation and trait evolution. For continuous characters, the root state was always set to 0. λ = speciation; μ = extinction; P_{ij} = transition rate; σ^2 = evolutionary variance; θ = optimum, α = rate of adaptation toward the optimum.

Data generation	**Model**	**Parameters**
Phylogeny	Birth-death	$\lambda = 1.0,\ \mu = 0.9$, extant tips $= 50, 100, 150$
Continuous traits	BM	$\sigma^2 = 0.25$
	$BM + trend$	$\sigma^2 = 0.25,\ \text{trend} = 0.3$
	OU	$\sigma^2 = 0.25,\ \theta = 0, \alpha = \frac{\ln(2)}{\text{tree height} \times 0.25}$
Discrete traits	$Fast$	$P_{ij} = 0.9$
	$Medium$	$P_{ij} = 0.1$
	$Slow$	$P_{ij} = 0.01$

2.2 Fossil sampling simulation

To explore the interaction of fossil sampling with estimation method on the disparity accuracy, we simulated fossil sampling using two different schemes. The first scheme, labelled 'tip-only', is that extinct tips were either retained ('fossilised') or pruned using a Bernoulli sampling method at five different probabilities: 0%, 5%, 15%, 50% and 100% (Figure 1: Step 3). This method meant each extinct tip had an equal probability of being 'fossilised' at each sampling level, irrespective of the total tree size.

To test whether the 'tip-only' fossil sampling scheme was biased by only sampling tips, we used the same Bernoulli sampling method but allowed nodes to be sampled as well as tips. This is labelled the 'tips+nodes' fossil sampling scheme. Given the computational expense of the analyses, we ran this alternative scheme on the 50 living tips (<500 total tips) trees only. Ancestral state estimation requires observed data to be coded as tips on a phylogeny, so sampled nodes were attached to the tree as tips using the `phytools::bind.tip` function (Revell 2012) with a non-zero branch length. The corresponding trait matrices were subset to match the tips remaining in each down-sampled tree, with all extant taxa sampled across all fossil sampling densities. From herein, these resulting downsampled trees and matrices are referred to as sampled-taxa.

2.3 Ancestral state estimation (ASE)

We estimated the ancestral states of the sampled-taxa using `dispRity::multi.ace` (Guillerme 2018), which is a wrapper for `ape::ace` (Paradis and Schliep 2019). We used four combinations of two approaches to ancestral state estimation to evaluate each method's accuracy in recovering true disparity values (Figure 1: Step 4). These were then compared to a null of bypassing ancestral state estimation whereby only sampled-taxa were used to generate trait spaces.

2.3.1 Pre-ordination estimation

Ancestral state estimation of trait data can be carried out at two points either side of the ordination step in the pipeline (Figure 1: Step 4; Lloyd 2018). Pre-ordination ancestral state estimation estimation involves the estimation of the raw trait values for each node. We performed pre-ordination estimation of continuous characters under a BM model, whereby ancestral state values lie within a multivariate normal distribution that is estimated given the observed trait data and phylogeny (Schluter et al. 1997; Rohlf 2001).

Pre-ordination estimation of discrete characters involves estimating the marginal likelihoods for each discrete state at each node, based on a maximum likelihood derived estimate of the Mk transition matrix and phylogeny given (Pagel 1999; Yang 2006). Given that the transition rates were set *a priori* during simulations, the equal-rates (ER) model was used to avoid model misspecification as a confounding factor on estimation accuracy (Keating 2023).

We constructed trait spaces for disparity estimation by ordinating the matrices of combined sampled-taxa and ancestral node states. Continuous data were ordinated using principal components analysis (PCA; `stats::prcomp`). Because PCA requires raw continuous data to calculate covariance, discrete characters were instead converted to distance matrices using the Maximum Observed Rescaled Distance metric (MORD) (Lloyd 2016) and ordinated using principal coordinates analyses (PCoA; `stats::cmdscale`) using the Cailliez correction to handle negative eigenvalues (Cailliez 1983).

2.3.2 Post-ordination estimation

Post-ordination estimation calculates the continuous PC scores across each axis of an ordination matrix for each ancestral node. For post-ordination ancestral state estimation, PCA and PCoA were applied prior to ancestral state estimation to the down-sampled continuous and discrete character matrices, respectively, to generate a set of sampled-taxa trait spaces. Ancestral node PC scores were then estimated under a Brownian motion model across all PC axes.

2.3.3 Uncertainty treatment

For each estimation method, we also compared point estimate and probabilistic approaches. The point estimate approach returned a fixed single value for each estimate. For continuous characters, the point estimate returned by `ape::ace` is the single maximum likelihood estimate of the ancestral state. For discrete characters, we used a strict majority rule with the state with the highest scaled likelihood selected at each node for each character. When state likelihoods were tied one state was selected at random.

The probabilistic approach differed in that uncertainty was explicitly accounted for by sampling multiple estimates rather than a single point. For continuous characters, we sampled each state across 100 matrices, with each estimate value lying within a normal distribution centred on the single point estimate and spread across the 95% confidence interval. Each discrete character state at each node was also sampled across 100 matrices according to their scaled

likelihoods. In a scenario that there is uncertainty in a binary character at a specific node, e.g., 0.55 for state 0 and 0.45 for state 1, this distribution approach results in a 55% probability of sampling state 0 in each matrix for the respective node (and 45% chance of sampling state 1). Both procedures were implemented using the `sample = 100` argument in `dispRity::multi.ace` (Guillerme 2018).

2.4 Disparity analyses

Our aim for this study was to understand how well ancestral state estimation can recover true properties of size and density in trait space occupancy, measured by disparity metrics (Guillerme et al. 2020b). Size-based metrics describe the volume of trait space the clade occupies (Guillerme et al. 2020b). To examine how well the size of the trait space was recovered, we calculated the sum of quantiles. The sum of quantiles measures the 95% interpercentile range on each PC axis and then sums these ranges. This metric has similar properties to the often used sum of ranges metric by describing the total trait space exploited (Foote 1992b), but with less sensitivity to outliers and sample size. For density, we calculated the sum of variances and mean pairwise distance (Foote 1991, 1992a). To quantify methodological performance, we calculated these disparity metrics for the ground-truth simulated trait spaces, the sampled-taxa only trait spaces, and the spaces generated by each ancestral state estimation approach so that relative disparity errors could be calculated (Figure 1 step 5). Given that the dimensionality of discrete trait spaces varied across fossil sampling densities within the same replicate, we also ran disparity calculations with dimensionality in the true and estimated trait spaces constrained to the minimum number of axes reached across all fossil sampling levels, i.e., for the 50 living-taxa trees this equalled 48 axes.

$$\text{Relative disparity error} = \frac{\text{Estimated disparity} - \text{True disparity}}{\text{True disparity}} \quad (2)$$

2.5 Statistical tests

The relative disparity errors were converted to absolute relative errors to measure the magnitude of deviation from the true disparity, regardless of direction. Due to the zero-inflated and right-skewed nature of the error distribution, we applied a log-transformation. Consequently, the more negative the log-error, the more accurate (closer to $log(0)$) the original estimation. Given the imbalance in sample sizes between the different uncertainty treatments (i.e., distribution approaches returned

100 disparity values per estimate, whereas point estimates returned one), we aggregated the distribution outputs by taking the median prior to analyses. We also performed analyses in which the distribution was retained, but each individual point returned by distributions was weighted 1/100th of the single point estimate.

We investigated the drivers of disparity estimation accuracy with linear-mixed models (LMM) using the `lme4` R package (Bates et al. 2015; Figure S5, S6) The log-transformed relative disparity error was treated as the response variable. Based on our experimental design, we fitted a four-way interaction between the estimation method, model of evolution, fossil sampling density and disparity metric. We also included tree size as a further non-interactive fixed effect after finding it had no significant interaction with the other predictors. To account for the variation with each simulation run, each tree replicate was included as a random intercept. We carried out *post-hoc* pairwise comparisons using estimated marginal means, adjusted using Tukey's honestly significant difference (HSD) (Lenth and Piaskowski 2025).

3 Results

The results reported herein are from the aggregated linear mixed-effects models (LMMs) from the tip-only fossil sampling simulation analysis retaining all PC axes in trait space. Results from the weighted distribution point LMMs were consistent with these findings (Figure S7, S8).

3.1 Continuous traits

3.1.1 Estimation methods

For continuous characters, the estimation method had a significant effect on disparity accuracy ($F_{4,66976} = 358,\ p < .001$; Figure 2). When averaged across all scenarios, ancestral state estimations improved estimates of disparity compared to excluding ancestral states. *Post-hoc* pairwise comparisons of estimated marginal means (EMM) revealed both pre- and post-ordination distributions of ancestral state estimations returned the lowest log disparity errors (Table S1). In contrast, point estimates performed worse overall than bypassing ancestral state estimation altogether.

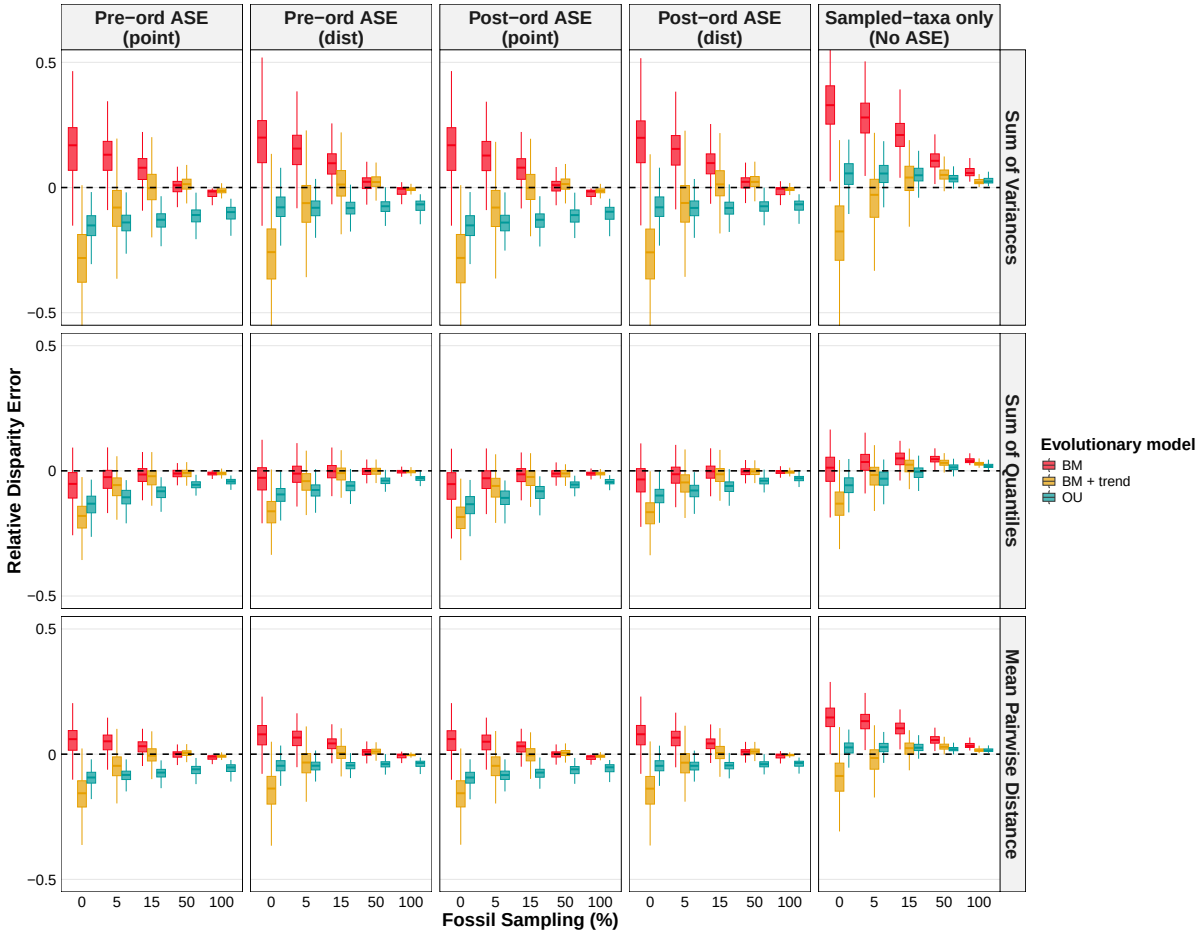


Figure 2: Relative disparity error returned from pre-ordination point ancestral state estimation (ASE), pre-ordination distribution ASE, post-ordination point ASE, post-ordination distribution ASE and no ASE. These disparity errors were calculated across three evolutionary models (Brownian motion [BM], BM+trend and OU), five simulated fossil sampling densities (100%, 50%, 15%, 5% and 0%) and three disparity metrics (sum of variances, sum of quantiles and mean pairwise distance). Values closest to 0 (dashed horizontal line) are most accurate; negative errors are under-estimates and positive are over-estimates.

3.1.2 Interaction effects

However, the pattern that ancestral state estimation methods improved estimates of disparity was context-specific (Figure 3). A three-way interaction between estimation method, fossil sampling and evolutionary model was significant ($F_{32,66976} = 60,\ p < .001$) and the *post-hoc* EMM can be used to identify the optimal method for models at each fossil sampling density (Table S2). For Brownian motion (BM) at 100% fossil sampling, the distribution estimations were most accurate. At 50% and below, methods of ancestral state estimation generally performed equally well and significantly better than excluding ancestral states. Under BM + trend, a similar pattern was followed at high fossil sampling densities, yet when fossil sampling dropped below 15% methods of ancestral state estimation performed worse than excluding ancestral states. In

contrast, under an OU model the exclusion of ancestral states was the best approach across all fossil sampling densities. Notably, we found that fossil sampling was the strongest driver of recovery of an accurate disparity value for continuous traits ($F_{4,66976} = 9169,\ p < .001$), as each increase in fossil sampling density resulted in a significant reduction in disparity error. While the disparity metric had a large effect on the overall estimation accuracy, it was a relatively lower driver of variation regarding its interaction with the estimation method ($F_{8,66976} = 7,\ p < .001$). Therefore, while estimation accuracy differed between metrics, the hierarchy between optimal methods was generally consistent across metrics.

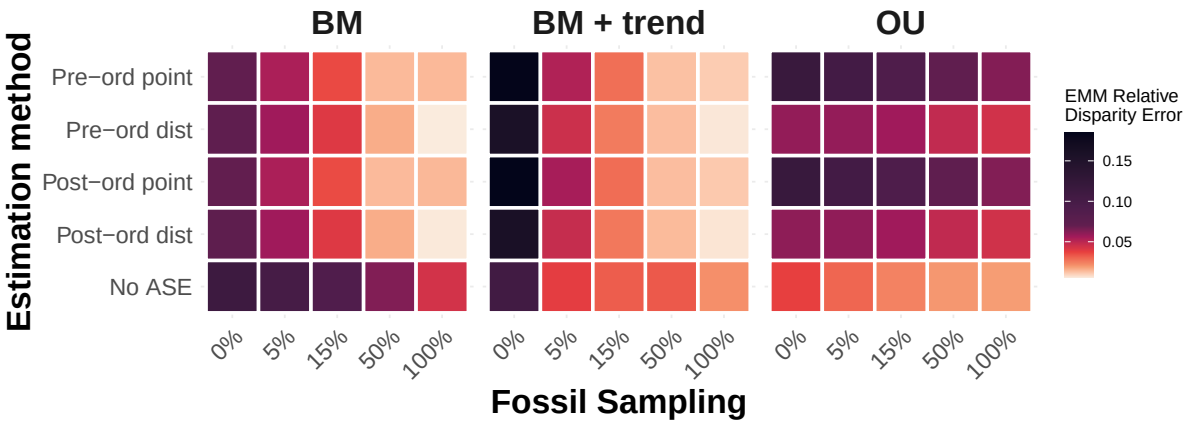


Figure 3: Heatmap presenting the continuous estimated marginal mean (EMM) relative disparity errors for every combination of estimation method, evolutionary model and simulated fossil sampling density. EMM were calculated from a linear mixed model with a three-way interaction term, averaged across disparity metrics.

3.1.3 Sensitivity tests

We found that the hierarchy of methods across continuous trait models and fossil sampling densities were largely congruent under the alternative 'tips+nodes' fossil sampling scheme, albeit we found that excluding ancestral states was marginally better performing when using the sum of quantiles metric on BM trait data (Figure S11). We also found our findings were robust to asymmetry in trees (Figure S9).

3.2 Discrete traits

3.2.1 Estimation methods

For discrete characters, the estimation method also had a significant effect on disparity accuracy ($F_{4,66976} = 10767,\ p < .001$; Figure 4). Similar to continuous traits, methods of ancestral state

estimation consistently improved disparity estimates compared to excluding ancestral states. However, the optimal type of estimation differed. *Post-hoc* pairwise comparisons of EMM revealed that pre-ordination point estimates returned the lowest log disparity errors overall. In contrast to continuous traits, distribution methods generally performed worse than pre-ordination point estimates on average, but post-ordination point estimates performed worse than excluding ancestral states (Table S3).

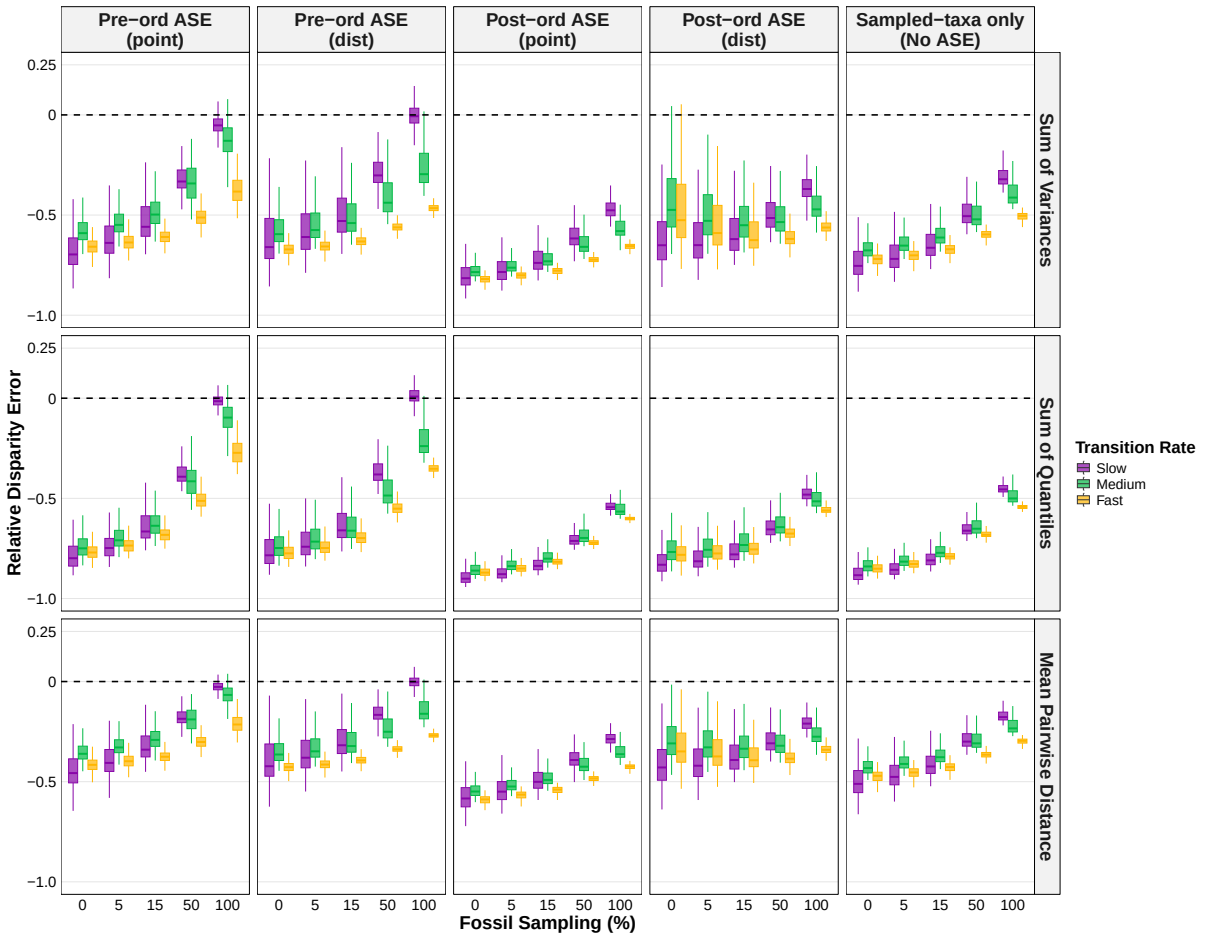


Figure 4: Relative disparity error returned from pre-ordination point ASE (ancestral state estimation), pre-ordination distribution ASE, post-ordination point ASE, post-ordination distribution ASE and no ASE. These disparity errors were calculated across three transition rates (fast [0.9], medium [0.1] and slow [0.01]), five simulated fossil sampling densities (100%, 50%, 15%, 5% and 0%) and three disparity metrics (sum of variances, sum of quantiles and mean pairwise distance). Values closest to 0 (dashed horizontal line) were most accurate; negative errors are under-estimates and positive are over-estimates.

3.2.2 Interaction effects

Similarly to continuous traits, the three-way interaction between estimation method, fossil sampling and transition rate explained much of the variation in the data ($F_{32,66976} = 588$, $p < .001$;

Table S4). The broad pattern was that at high fossil sampling densities, pre-ordination point estimates were the best method, whereas at lower fossil sampling densities both pre- and post-ordination distributions returned the most accurate disparity values (Figure 5). This pattern was transition rate-dependent, however, as under a slow rate the pre-ordination distribution method was the best performing method across all fossil sampling densities. At fast and medium rates under low fossil sampling ($< 15\%$), the post-ordination distribution method returned the lowest disparity errors. Notably, methods of ancestral state estimation outperformed their exclusion across all combinations of transition rate, fossil sampling density and disparity metric. Again, while the disparity metric was a significant driver to overall accuracy, it had little effect on the hierarchy of methods defined by the interaction between transition rate and fossil sampling density, signified by a weak four-way interaction ($F_{64,66976} = 10,\ p < .001$).

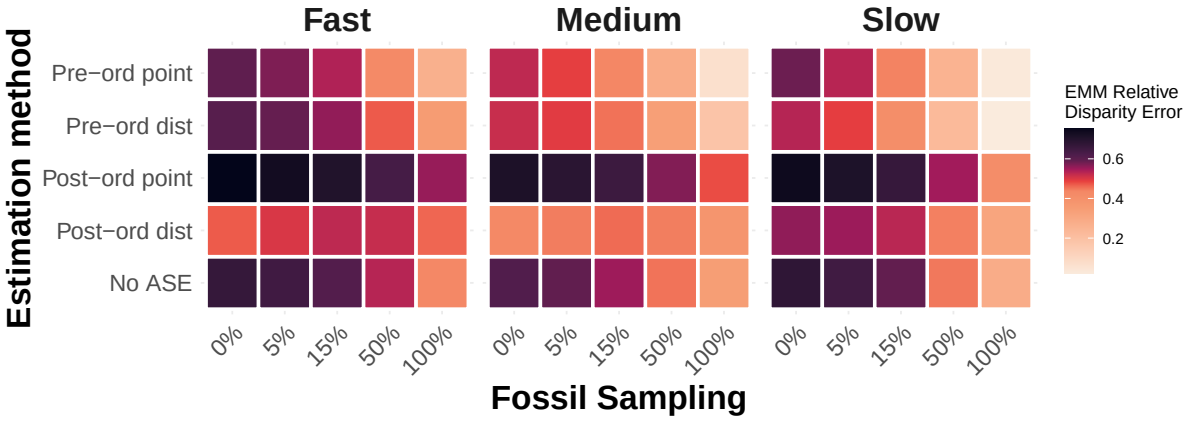


Figure 5: Heatmap showing the discrete estimated marginal mean (EMM) relative disparity errors for all combinations of estimation method, transition rate and simulated fossil sampling density. EMM were calculated from a linear mixed model with a three-way interaction, averaged across disparity metrics.

3.2.3 Sensitivity tests

Under the alternative 'tips+nodes' fossil sampling scheme, we found that, generally, the hierarchy of methods remained consistent with our main findings up to 50% fossil sampling density (Figure S12, Figure S13). Beyond 50% fossil sampling, we overestimate disparity with pre-ordination methods, because we now have twice as many points as the true trait space because we are estimating ancestors that did not exist. Essentially, under a 'tips+nodes' sampling scheme ancestral state estimation becomes redundant beyond 50% fossil sampling because all unsampled taxa are accounted for as ancestral nodes. Our findings also proved robust under asymmetrical

trees (Figure S10).

When we account for unequal dimensionality, we found virtually the same hierarchy of estimation methods as when we preserved the full dimensionality of the trait space (Figure 6), but note that the extent at which pre-ordination estimation methods were advantageous over excluding ancestral state estimations was slightly diminished.

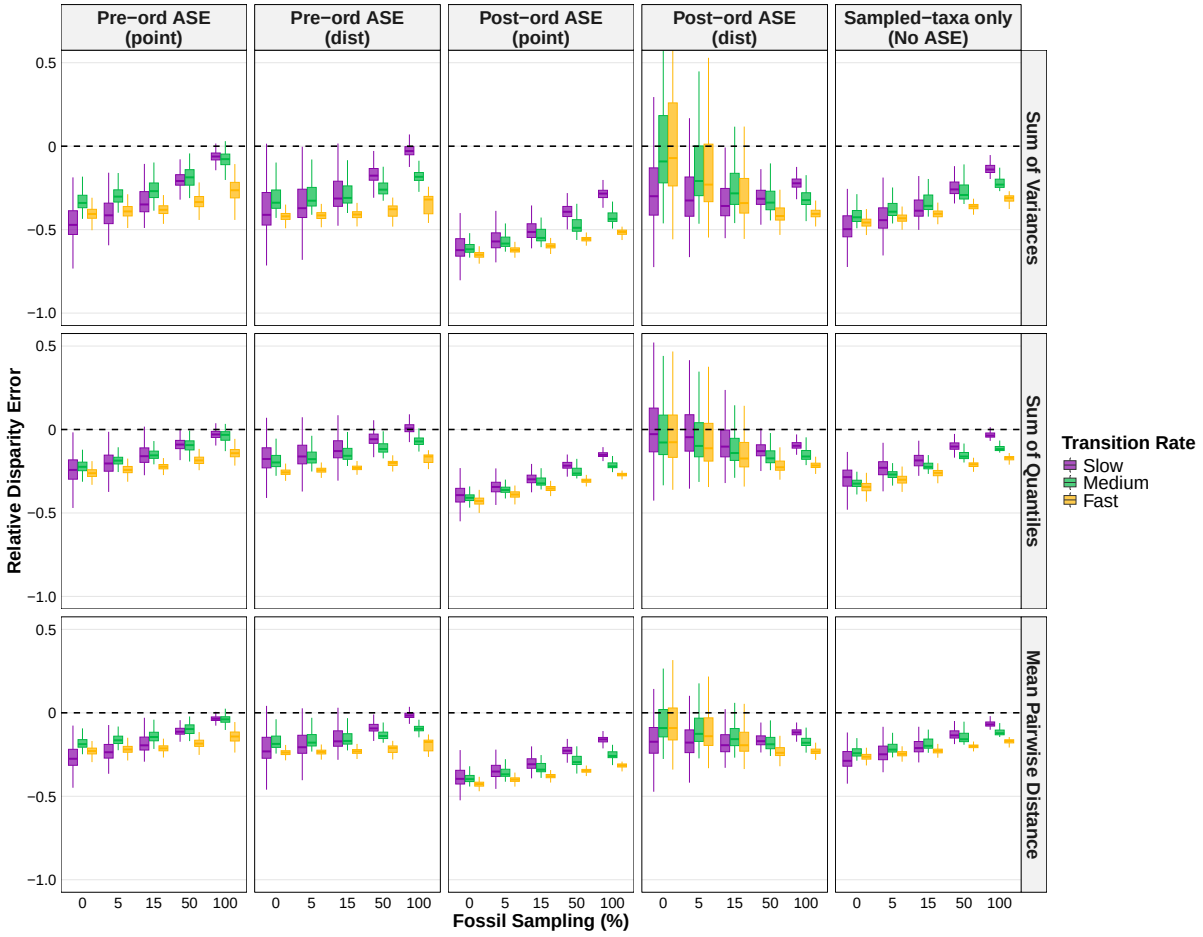


Figure 6: **fixed axes** Relative disparity error returned from pre-ordination point ASE (ancestral state estimation), pre-ordination distribution ASE, post-ordination point ASE, post-ordination distribution ASE and no ASE. These disparity errors were calculated across three transition rates (fast [0.9], medium [0.1] and slow [0.01]), five simulated fossil sampling densities (100%, 50%, 15%, 5% and 0%) and three disparity metrics (sum of variances, sum of quantiles and mean pairwise distance). Values closest to 0 (dashed horizontal line) are most accurate; negative errors are under-estimates and positive are over-estimates.

4 Discussion

Our results suggest that, generally, the inclusion of ancestral states improves the recovery of true disparity, defined here as the total region in trait space occupied by all members of a clade that have ever lived. However, we found this to be context-specific, particularly dependent on fossil

sampling and model of evolution. Overall, we recommend that ancestral state estimations can consistently improve estimations of disparity derived from discrete characters, but recommend caution for continuous characters.

4.1 Pre- versus post-ordination estimation

For continuous characters, we found virtually no discrepancy between the effect of pre- and post-ordination estimation of ancestral states on disparity. Given that continuous traits exist within a Euclidean space, the linear transformations of both PCA and ancestral state estimation under Brownian motion (BM) mean that estimated node states exist within the same relative positions to the observed tip data for both pre- and post-ordination estimation. Note, however, that our simulated traits were uncorrelated. In the case of non-orthogonality between traits, it is likely that the position of ancestral state estimation in the disparity pipeline will have some effect, although this was not tested here.

For discrete traits, the position of ancestral state estimation relative to ordination during the disparity pipeline (Figure 1) was a key driver in accurate recovery of the true disparity. Overall, pre-ordination estimation led to more accurate disparity estimations than post-ordination estimation, a finding consistent with previous work (Lloyd 2018). Post-ordination estimation is ultimately limited by the definition of the resulting trait space. Axes of the trait space are treated as continuous traits evolving under BM, so, given the assumptions of BM, ancestral nodes are inevitably constrained within the bounds per-PC axis defined by the observed tip data (Stayton 2015; Lloyd 2018; Gerber 2019). This forces the implicit assumption that our observed trait data accounts for the extremes of morphological diversity for the clade of interest, and that past diversity is only a subset of sampled diversity. Disparity is therefore underestimated, particularly with range-based metrics such as sum of quantiles (Figure 4). Pre-ordination estimation does not encounter these same limitations. A discrete trait matrix exists within a hypercube, where each vertex corresponds to a unique combination of traits. When ancestral states are estimated prior to ordination, there is no constraint preventing an ancestor from occupying a legitimate and theoretically possible vertex that lies peripheral to the observed data (Gerber 2019). These ancestral nodes that lie outside the multidimensional convex hull defined by the tip data can inflate disparity relative to sampled-taxa only disparity (Brusatte et al. 2011). Similarly, authors have previously noted that inclusion of ancestral nodes can inflate disparity by (nearly) doubling the number of data points included in the trait space (Brusatte et al. 2011), raising questions

about whether such increases reflect genuine biological signal. However, our findings demonstrate that the trait space expansion from the inclusion of ancestors, particularly when fossil sampling density is high, does converge on the true disparity of the clade under pre-ordination estimation. As Figure 4 shows, underestimation of disparity is ubiquitous to all discrete character estimation methods, which has previously been linked to character exhaustion (Erwin 2007). Ultimately, pre-ordination discrete estimations are less susceptible to extreme underestimation because ancestral nodes are not bound to a weighted average of the tip data in the way post-ordination estimation is.

4.2 Treatment of uncertainty

The choice of estimation method for continuous characters was largely driven by the treatment of uncertainty. At high fossil sampling densities, incorporating uncertainty through the use of distributions performed better than point estimates (Figure 2; Figure 3). This is due to the tendency for point estimates to smooth over evolutionary history (Boyko and O'Meara 2024; Boyko et al. 2026), underestimating the morphological variation of ancestral taxa. However, in contrast to discrete traits, continuous estimation distributions are unbounded and thus sensitive to data quality. Therefore, the variation introduced from sampling across a distribution was dependent on the density of fossils; at high fossil density, confidence intervals are constrained due to lower uncertainty in estimates. As fossil sampling decreases, these confidence intervals are widened and more noise is introduced as a result of using distribution (Schluter et al. 1997; Martins 1999), negating the advantage observed at higher fossil sampling.

For discrete characters, we found an inverse pattern in which distribution methods were optimal at low fossil sampling densities and point estimates were optimal at high fossil sampling densities. This is likely due to the bounded nature of discrete traits; even with minimal information, the disparity error that can be returned is bounded by the number of states present at each node, which contrasts the effectively infinite error that can be returned by continuous characters. Regarding pre-ordination estimations, our results reveal two issues relating to uncertainty. At low fossil sampling or slow transition rates, the predominant issue facing the recovery of a true trait space is the lack of data. For fossil sampling, the lack of evidence in deeper time means our ancestral state estimates, and subsequent trait space pattern, are biased by the living tip data. In the case of slow rates, we face a paradoxical situation in which the root state can be easily estimated, yet few transitions make the rate of change hard to estimate (Gascuel and

Steel 2020). Using the hypercube analogy again, the state scaled likelihoods at each node under slow rates or low/absent fossils will be biased against transitions. Therefore, because point estimations essentially recover the most parsimonious evolutionary history, ancestral nodes will be constrained within a certain region of vertices in the trait space. In contrast, pre-ordination distribution estimations allow the exploration of other vertices by sampling across alternative discrete state combinations. In the case of slow rates and low fossil sampling, this was found to improve disparity estimations (Figure 5). The flip side of this is the issue of saturated data, where the paradox is reversed and instead, we have an abundance of information, thus transition rates are easy to estimate, yet due to the rate of change most nodes are left with inconclusive scaled likelihoods. Under the fast rates simulated in this study, character transitions are saturated, so there is very little phylogenetic signal. The use of distributions amplifies the noise in this scenario, thus point estimates can be used to filter out the noise which is why they outperform distributions at medium/fast rates (Figure 5).

4.3 Model misspecification and fossil inclusion

As expected, we found that the differences in true and simulated disparity across models of evolution reflect model misspecification in ancestral state estimation. The inclusion of ancestral state estimations generally recovered a more accurate trait space than excluding ancestral states when BM was the simulated model. This was an expected finding, given that ancestral state estimations of continuous characters were modelled under BM. The assumptions of BM include a fixed mean and trait variance increasing linearly through time (Felsenstein 1985; O'Meara et al. 2006). We found that when these assumptions of BM were violated, such as under BM + trend and OU models, disparity accuracy returned by methods with ancestral state estimation decreased. For example, under low/absent fossil sampling, methods of ancestral state estimation performed worse than bypassing ancestral state estimation for BM + trend. In this case, disparity estimates tended to be underestimated, because state estimations will be intermediates of the observed tip data, absent to the directional trend that occurred and thus missing drifted regions within trait space. Our results suggest a threshold between 5-15% fossil sampling where ancestral state estimation becomes optimal over its exclusion in disparity (Figure 3). This finding supports previous work that highlights the critical impact of fossil inclusion when directional trends are present in the trait data (Finarelli and Flynn 2006; Albert et al. 2009; Slater et al. 2012; Finarelli and Goswami 2013; Gearty et al. 2024).

Similarly, calculations of disparity derived from OU simulations were also underestimated when using ancestral state estimations. The OU process is characterised by a stationary variance, which contrasts the BM assumption of an increase in variance through time. Therefore, because BM estimates internal nodes as a weighted average of the observed tip data, the ancestral nodes are compressed towards the centroid of the trait space. This explains why, when model misspecification is fundamentally incorrect, bypassing ancestral state estimation outperforms the use of ancestral state estimation, even with 100% fossil sampling. We therefore encourage model-fitting prior to performing ancestral state estimation to avoid model misspecification; several `R` packages allow model-fitting and estimation of continuous traits under an OU model (Revell 2012; Pennell et al. 2014; Clavel et al. 2015). Note that empirical studies of disparity tend to focus on clades that are suspected to have undergone major evolutionary shifts, such as adaptive radiations (Ruta et al. 2013; Stroud and Losos 2016) or mass extinctions (Bapst et al. 2012; Halliday and Goswami 2016; Liu et al. 2024), which are often associated with shifts in evolutionary rates or models. The effect of such shifts or heterogeneity in the context of the aims of this paper remains to be tested; previous work has found that rate changes are hard to detect and can lead to erroneous inferences in single-trait ancestral state estimations (King and Lee 2015; Chira and Thomas 2016; Harrington and Reeder 2017; but see Reyes et al. 2018; Muñiz-Trejo et al. 2025). Several recent methodological developments have aimed to accommodate for rate heterogeneity in both continuous (Elliot and Mooers 2014; Smaers et al. 2016; Castiglione et al. 2020) and discrete ancestral state estimation (Boyko and Beaulieu 2021; Revell and Harmon 2024) and so the testing of these methods in the context of disparity is a potential future avenue of research.

4.4 Simulation sensitivity tests & limitations

Simulations are increasingly used across computational evolutionary biology to evaluate methodological approaches (Lotterhos et al. 2022). However, given the complexity of morphological evolution across clades, generalising simulation frameworks is an endeavour that offsets empirical and biological realism (Goloboff et al. 2018; O'Reilly et al. 2018). Where tested, our conclusions generally hold under our simulation assumptions. For empirical realism, the tip-only fossil sampling scheme is the more defensible default for our main analyses. A fossil cannot be identified as an ancestor or descendent when found, and fossil taxa are added to trees as tips, so this scheme emulates the typical procedure. However, the tip-only scheme excludes taxa that are ancestral

to other sampled taxa, i.e., ancestral nodes are not sampled, despite evidence that ancestors being may be at least as likely to be sampled as descendent taxa (Foote 1996; Parins-Fukuchi 2026; note that our simulations assume a bifurcating dendogramic evolutionary history, thus if according to branch lengths internal branches are no more likely to be sampled than terminal branches, though this assumption changes if budding is considered as it is in the studies cited). The alternative 'tips+nodes' sampling scheme highlighted where these assumptions may have introduced bias. For discrete characters, the tip-only scheme appears to introduce little bias. Tips and nodes did not differ in their pairwise distances (Figure S14), so excluding ancestors did not remove a distinct part of the trait space. This is consistent with character exhaustion in bounded discrete characters (Wagner 2000; Erwin 2007); once most character states have been expended, further change repeats existing pairwise distances, thus tips do not accumulate any more distinctness than internal nodes. This trait space saturation is reached faster as transition rates increase. In contrast, continuous characters are unbounded. Under BM, variance increases with distance from the root, so tips are the most morphologically distinct taxa (Figure S15). This explains why the tip-only sampling scheme overestimates BM sum of variances at low fossil sampling when using ancestral state estimations, whereas the 'tips+nodes' scheme instead underestimated sum of variances. Our conclusions for both discrete and continuous characters are robust under both sampling schemes, but we note that the scheme does have a noticeable effect on continuous results.

Given that empirical phylogenies tend to be asymmetrical (Mooers and Heard 1997), we tested for the effect of asymmetry on our results (**??**) and found congruent patterns in estimation method hierarchy across treatments to our main findings. Raw ancestral state estimation accuracy has been shown to increase as tree asymmetry increases, since asymmetrical trees have fewer nested internal nodes, although longer branch lengths can increase uncertainty in estimations (Keating 2023). We observed this exact pattern, with the generally lower disparity error across asymmetry trees, albeit with wider error within pre-ordination estimation methods (**??**).

Our findings also depend on our assumption of bifurcating cladogenesis, in which ancestors will split into two daughter species. Under budding cladogenesis, progenitors are able to persist alongside their descendents, thus long-lived lineages are more likely to be ancestral and sampled (Parins-Fukuchi 2026). Ancestral state estimation (and phylogenetic comparative methods in general) do not accommodate budding (Caetano and Quental 2023) and so whether our results hold when ancestors persist and overlap with their descendents remains to be tested.

Several assumptions remain untested. We did not model phylogenetic uncertainty; Furthermore, fossil preservation was an assumption we did not model. While we varied fossil sampling density, every sampled fossil had complete character data. Fossil preservation, in other words character matrix completeness, is uneven across clades and time and can be concentrated at lagerstätten, which can distort inferences from disparity analyses (Butler et al. 2012; Flannery Sutherland et al. 2019) and also phylogenetic inference (Sansom et al. 2010; Woolley et al. 2024) How this exceptional preservation can bias the scaled likelihoods of states at ancestral nodes and affect disparity downstream remains to be tested.

Similarly, fossil degradation can affect inference; How could Lagerstätten effects be changed with ancestral state estimation? Given that these factors have not been accounted for, we recommend that phylogenetic corrections to disparity curves do not replace sampled-taxa only curves, but that both are used complimentary in analyses.

4.5 Conclusions and recommendations

Our findings present a convincing case for the use of ancestral state estimation in disparity analyses, a method that has often been neglected in disparity workflows. Furthermore, our results issue a stark reminder of the essential need for fossils during macroevolutionary workflows. The estimated proportion of fossils sampled across empirical datasets is typically less than 5% (Raup 1976), which ultimately limits our inferences into macroevolutionary patterns. However, in the case of discrete traits, the incorporation of ancestral states is more optimal than their exclusion across all fossil sampling densities, and the same is true of continuous traits when the correct model is used for estimation. Based on these findings, we can recommend the following guidelines for the use of ancestral state estimation in disparity analyses: 1) for continuous traits, perform model testing prior to running ancestral state estimation to infer the likely evolutionary model parameters, and then estimate ancestral states under this model, using the distributions of estimates in disparity analyses; 2) for fast/medium rate discrete traits, pre-ordination point estimates should be used when fossil sampling is suspected to be less than 15% and post-ordination distribution when fossil sampling is lower than this. Pre-ordination distribution should be used when transition rates are slow across all fossil sampling densities.

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Author Contributions

Conceptualisation CN Scutt (CNS), GH Thomas (GHT), N Cooper (NC), T Guillerme (TG); **Data curation** CNS; **Formal analysis** CNS; **Funding acquisition** TG; **Investigation** CNS, TG; **Methodology** CNS, TG; **Project administration** CNS, GHT, NC, TG; **Resources** CNS, TG; **Software** CNS, TG; **Supervision** GHT, NC, TG; **Validation** CNS; **Visualisaton** CNS; **Writing - original draft** CNS; **Writing - review & editing** CNS, GHT, NC, TG.

Data Availability

The code to reproduce the analyses is available here `https://github.com/calebowski/acedisparity` (Scutt and Guillerme 2026). Data is available at `https://doi.org/10.5061/dryad.280gb5n4p`.

Competing Interests

We declare no conflict of interest.

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